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PALEOMAGNETISM OF SILURIAN-ORDOVICIAN SEDIMENTS FROM THE VALLEY AND RIDGE PROVINCE, NORTHWEST GEORGIA

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Abstract. The remanent magnetism (RM) for 93 specimens has been determined in 34 independently oriented samples from 20 sites at Ringgold Gap, Georgia, in the Valley and Ridge Province of the Southern Appalachians. Lithologies sampled include Middle Ordovician through Silurian fossiliferous limestones and hematitic sandstones and siltstones. The section is confined within a single thrust slice. Thermal demagnetization results indicate that hematite is the dominant RM carrier although detrital magnetite has also been identified optically in some specimens.

Both normal and reversed polarities are present and the resulting paleopole is 28°N, 142°E (α_{95} = 7.3°). This pole falls within the cluster of published Lower Paleozoic Valley and Ridge poles but rather than clarifying the distribution of poles it augments the scatter. Three factors are thought to contribute to this scatter: (1) unresolved components of rotation, possibly introduced during folding and thrust faulting, which result in declination anomalies, (2) true vertical axis tectonic rotations during deformation, and (3) insufficient knowledge of the timing of magnetization. In light of these uncertainties, caution should be exercised in constructing a Lower Paleozoic apparent polar wander path with paleopoles from the Valley and Ridge Province.

Introduction

Paleomagnetic pole positions for the Lower Paleozoic of North America are limited, and show considerable scatter (Buchan and Hodych, 1982; Irving, 1979; Watts et al., 1980). Explanations for this scatter include remagnetization (Van der Voo and French, 1977), tectonic rotations (Gillett, 1982), and/or translations (Irving, 1979; Kent and Opdyke, 1978), and rapid Apparent Polar Wander (Gillett, 1982). As the quantity and quality of paleomagnetic data for Lower Paleozoic North America increases, our understanding of problems inherent to the data improves. This study was designed to supplement the Lower Paleozoic data base by establishing an additional reliable paleopole for the Early Silurian and Late Ordovician. A number of previous Ordovician-Silurian paleomagnetic studies have been conducted in the Valley and Ridge Province of the central and southern Appalachian Mountains (Van der Voo and French. 1977; Gillett, 1982; Watts and Van der Voo, 1979; Roy et al., 1967), but considerable scatter is apparent in the data.

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Geologic Setting

The study area is located in the Valley and Ridge province of northwest Georgia. A deep road cut (the Ringgold Gap) bordering Interstate 75 exposes ~180 meters of folded Middle to Upper Ordovician and Silurian sediments. The cut is through Taylor Ridge, situated in a thrust slice between the Clinchport thrust (to the east) and the Kingston thrust (to the west). Lithologies include fossiliferous limestones, calcareous mudstones, hematitic sandstones and siltstones, dolomitic siltstones and shales, and sandstones. The mean orientation of bedding for the section has a strike of N20E dipping to the SE at ~14° and the variability from site to site is low.

The stratigraphy of the sequence has been studied by numerous workers and is well known (Chowns and McKinney, 1980; Rindsberg, 1982; Milici and Wedow, 1977; Drahovzal and Neathery, 1971). The location of the Silurian-Ordovician boundary in the section has been determined based on paleontologic criteria utilizing lithologic characteristics as additional criteria. Some uncertainty remains as to the exact location of the boundary. A distinctive two foot thick sandstone (bed 35 of Chowns and McKinney [1980]) marks the location of the boundary - although some uncertainty remains as to whether the boundary lies at the base or top of this sandstone unit (Chowns, written communication, 1982).

The Paleozoic sediments of the Valley and Ridge province are folded and extensively thrust faulted. Although the southern Appalachians have been affected by three orogenic events which include the Taconian Orogeny (~480-450 Ma.), the Acadian Orogeny (~380 Ma.), and finally, the Alleghanian Orogeny (~300-250 Ma.), the folding and thrust faulting in the Valley and Ridge is thought to be the result of Alleghanian deformation.

Methods

Samples were collected with a portable drill and as oriented hand samples. Hand samples were taken where the exposed rock was too friable or fractured to permit drilling. A total of 20 sites were sampled over the ~180 meters of exposed section. Each site was confined within a discrete sedimentary bed. These sites included samples from seven drilled sites and thirteen oriented hand samples from 13 different sedimentary units.

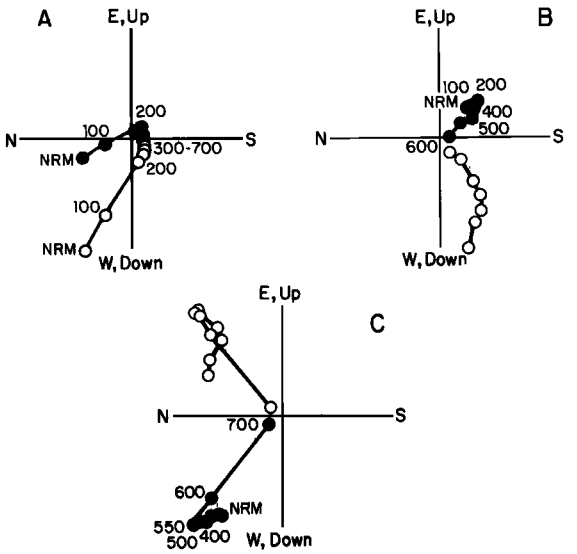


Fig. 1. Representative normalized orthogonal thermal demagnetization vector diagrams (Zijderveld, 1967). A: Sample RG-24-01; B: Sample RG-22-2B; C: Sample RG-19-05. Open Circles: vertical plane; Closed circles: horizontal plane. Data before structural correction.

The strike and dip of block samples and of the bedding plane at each site were measured using a Brunton compass. Structural corrections were performed for each site based on the assumption of bedding plane horizontality.

Ninety-three ~2.4 cm diameter cores, drilled in the field or in the laboratory from oriented hand samples, were cut to ~2.1 cm lengths. The RM was measured on a Schonstedt spinner magnetometer and an ScT cryogenic magnetometer. Progressive alternating field (a.f.) and thermal demagnetization was carried out on selected pilot samples. Based on linear trends visually apparent in the vector decay plots, all remaining samples were thermally demagnetized at 100°C increments to 500°C, then at 50°C increments to 700°C.

Results

Representative normalized orthogonal thermal demagnetization diagrams (Zijderveld, 1967) are shown in Fig. 1. In all cases apparently secondary components of magnetization were removed by 500°C to 600°C. RM means were derived by kappa maximization (Watkins, et al., 1972) using only 500, 550, 600, and 650°C thermal demagnetization data. Table 1 summarizes the site mean RM results after demagnetization and tilt correction. Sites denoted by an asterisk are those sites considered unreliable and have not been used in any subsequent paleopole calculations. Zijderveld (1967) vector diagrams for the samples from these sites either do not show stable components of magnetization, or the within site consistency is poor. Although some of the mean directions for unreliable sites fall relatively close to those from reliable sites, they have still been rejected. Two sites are also rejected based on anomalous polarities (25 or 26). These may be transitional directions. Fig. 2 shows the site mean RM directions after elimination of anomalous data and after structural correction, assuming originally horizontal bedding surfaces.

Discussion

The characteristic magnetization determined in this suite of samples is believed to reside dominately in hematite as a chemical RM (CRM). The presence and dominance of hematite in many samples is indicated by a red coloration and by a.f. demagnetization conducted on selected pilot samples, where even peak levels of 100 mT had little effect on the RM magnetization. This suggests that magnetite is not the dominant carrier of the RM in these samples although it is present optically in many samples examined and appears to be detrital in character.

Remanent Magnetism (RM)

Stepwise thermal demagnetization has revealed two components of magnetization (Fig. 1). The first, a low blocking temperature (T_B) component was removed by demagnetization at temperatures between 300 and 500°C (Fig. 1A-B). The low T_B for this RM component suggests that it may be a viscous RM (VRM) or possibly a low temperature overprint. Its direction is variable between sites. The second component of magnetization showed significantly higher blocking temperatures (Fig. 1C). In several samples this component was stable to nearly 700°C and decays linearly towards the origin on Zijderveld (1967) diagrams, indicative of univectorial behavior. The higher temperature component is interpreted as the characteristic magnetization.

The within-site precision for most sites is good (Table 1). In addition to good within site precision the 13 stable sites are well grouped, yielding good within section precision. The characteristic magnetization shows both normal and reversed polarities with typical antipodal directions. Only two sites (25 and 26) deviate from these well defined polarities (Fig. 2) and may be transitional or overprinted. They were not used in calculating the paleopole position reported here.

Timing of RM Acquisition

Although the timing of the acquisition of the characteristic magnetization can not be unequivocally determined, we estimate that the RM was

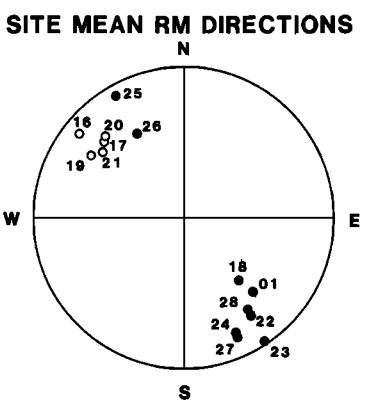


Fig. 2. Equal area plot of site mean characteristic magnetization directions, after rejection of anomalous data and structural correction. Points on lower hemisphere are filled circles. Upper hemisphere points are open circles.

Table 1. Site mean remanent magnetic results. N_1 is the number of samples drilled; N_2 is the number of samples used in calculating the site mean; D, I are declination and inclination in degrees after structural correction; P is the polarity, N being normal based on low inclination northwesterly directions, R reversed based on low inclination southeasterly directions; K is an estimate of precision, α_{95} cone of confidence (Fisher, 1953). An * after the site number denotes an unreliable site based on low K (<10) or anomalous inclinations (25 and 26) was not used in calculating the overall mean directions or paleopole reported. The Ordovician-Silurian boundary lies between sites 24 and 25, age increasing from top to bottom in Table 1.

Site	N_1/N_2	D	$\bar{I}$	$\bar{P}$	k	α_{95}
01	3/3	135.6	32.8	R	165.5	9.6
02*	4/4	82.4	33.0	R	6.2	40.1
03*	3/3	335.2	-16.2	N	1.4	180
04*	3/3	318.0	45.2	R	1.3	180
05*	3/3	221.2	1.5	-	1.2	180
06*	2/2	338.1	-10.1	N	2.8	180
07*	2/2	276.8	- 4.5	N	102.8	24.9
17	4/3	313.2	-28.5	N	20.0	28.3
16	6/5	308.8	-12.5	N	91.9	8.0
18	6/6	137.2	44.1	R	153.7	5.4
19	6/6	304.4	-26.7	N	73.9	7.8
20	6/6	315.7	-26.7	N	68.5	8.2
21	5/4	309.1	-31.8	N	64.9	11.5
22	6/6	144.4	21.4	R	29.6	12.6
23*	6/6	145.3	2.1	R	4.0	38.3
24	5/5	154.5	18.3	R	11.8	23.3
25*	6/5	330.6	9.5	N	33.8	13.3
26*	5/4	332.1	36.5	N	12.7	26.8
27	6/6	154.9	15.0	R	131.7	5.9
28	6/6	143.6	26.3	R	63.9	8.4
Means before structural correction						
R	6	150.3	35.7	R	35.4	11.4
N	5	314.7	-40.3	N	90.0	8.1
Means after structural correction						
R	6	145.6	26.5	R	39.2	10.8
N	5	310.2	-25.3	N	93.5	8.0

acquired shortly after deposition. Our reasoning is based on two independent lines of evidence. First, site mean directions are in reasonable agreement with other established Silurian and Ordovician directions. This suggests that an RM was acquired shortly after deposition. Second, fine grained apparently detrital magnetite is present optically and appears to be the dominant RM carrier in some samples. For example in at least 2 cases (sites 22 and 24), the RM directions begin to scatter with thermal demagnetization above 500°C and all but 3 sites (16, 21, and 23) show the greatest within site RM precision at thermal demagnetization temperatures below 600°C, suggesting magnetite blocking temperatures. The directions in these samples do not differ from those samples which appear to be dominated by hematite. Assuming that the fine grained magnetite is essentially primary, then the close agreement of directions residing in hematitic samples indicates that the hematite CRM was acquired during or very shortly after deposition or during lithification.

Even though it is presumed that the RM was acquired shortly after deposition, the possibility of remagnetization during later burial, heating, folding and thrusting must be considered. Even though a low blocking temperature component of magnetization is present in sediments at Ringgold Gap it is unblocked with demagnetization temperatures of 500-550°C. A characteristic magnetization is evident at greater temperatures. Pullaiah et al. (1975) have shown that a thermal overprinting event with temperatures significantly >300°C would be required to unblock such an RM. In turn, well established Late Carboniferous and Permian directions for North America (French and Van der Voo, 1979) would be expected if Alleghanian remagnetization had occurred. Independently, conodont samples from the Ringgold Gap section record a Color Alteration Index (CAI) of 3 (Epstein et al., 1977), indicating a relatively low thermal overprint for the section of ~150°C. We conclude from our demagnetization data that this value is probably too low by at least a factor of 2.

Models for the acquisition of RM in red beds show large disparities in the timing of remanence acquisition (Larson et al., 1982; Channell et al., 1982). However, samples with possibly primary magnetite yield the same RM directions suggesting essentially co-eval directions.

Channell et al. (1982) have proposed that diagenetic hematite growth in limestones from Gubbio, Italy, takes place within ~100,000 years after deposition. If this model of diagenetic hematite growth from a geothitic percursor is invoked for the origin of the magnetization at Ringgold Gap, the timing of RM acquisition may be further constrained. This model, however, cannot completely account for the large amount of hematite that exists in the Ringgold Gap section.

Larson et al. (1982) have suggested that the magnetization of the Moenkopi Formation results from a hematite CRM acquired by intrastratal alteration over periods as long as tens of millions of years. They suggest that the authigenic growth of hematite results from alteration of iron bearing detrital minerals by intrastratal solutions. Although this or a similar process is probably important in the growth of hematite at Ringgold Gap the duration of this process as suggested by Larson et al. (1982) is considered unlikely in view of the close correspondence of the Ringgold paleopole with Ordovician-Silurian poles, and rock magnetic data reported above.

Paleopoles have been calculated for each site mean direction and from the mean direction of all normal and all reversed sites (Table 1). The difference between the two values is small and essentially antipodal. The mean paleopole calculated using 11 site mean paleopoles measured for the Ringgold section is 27.7°N, 141.6°E, α_{95} = 7.3. This paleopole, as well as other Ordovician and Silurian Southern Appalachian poles is plotted in Fig. 3. Our paleopole is very similar to the middle Ordovician pole reported by Watts and Van der Voo (1979). However, other Ordovician and Silurian poles from the area represent considerable longitudinally distributed scatter (Van der Voo and French, 1977; Watts and Van der Voo, 1979), and the poles do not follow any obvious age dependent progression (Fig. 3). These differences may result from a number of factors, such as plate movements or unresolved tectonic rotations resulting either from more than one phase of noncoaxial tilting or vertical axis rotation (MacDonald, 1980), possibly developed during thrusting in the Central and Southern Appalachians.

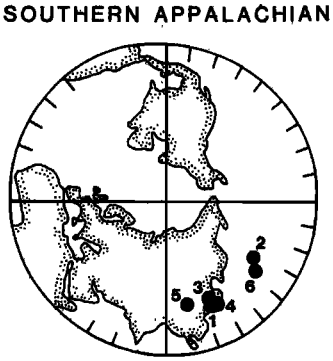


Fig. 3. North polar equal area plot of published paleomagnetic poles for the Ordovician and Silurian of the Valley and Ridge Province. Pole numbers correspond to Table 1. Poles are from Watts and Van der Voo, 1979 (#1, 2 = Middle Ordovian poles); Van der Voo and French, 1977 (#3 = Late Ordovician, #4 = Middle Silurian poles; Roy et al., 1967 (#5 = Upper Silurian pole); and this study (6).

Conclusion

The characteristic magnetization directions determined in sediments from Ringgold Gap, Georgia are in good agreement with other Ordovician-Silurian directions. The RM is inferred to reside mainly in hematite as a CRM, which acquired a magnetization shortly after deposition. The paleopole calculated from the RM directions is ~28°N, ~142°E and lies very close to a middle Ordovician Valley and Ridge paleopole (Watts and Van der Voo, 1979). Although scatter is apparent in the Valley and Ridge paleopoles the discrepancies are confined to variations in longitude. Therefore, we attribute the scatter seen in the Ordovician-Silurian poles to slight unresolved tectonic rotation or tilt in the section.

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